Quant Interview Problem Book

1000+ Problems with Worked Solutions

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Problem 2.55 (Differential equations (first-order linear ODE)). *Solve the initial value problem* $y'(t) + 0.5y(t) = 3$ *with* $y(0) = -1$.

Solution. This is a first-order linear ODE. Use an integrating factor.
Step 1 (integrating factor). The equation is

$$y'(t) + ay(t) = b, \qquad a = 0.5,\ b = 3.$$

The integrating factor is $\mu(t) = e^{at}$. Multiply both sides:

$$e^{at}y'(t) + ae^{at}y(t) = be^{at}.$$

The left-hand side is a derivative:

$$\frac{d}{dt}\Big(e^{at}y(t)\Big) = be^{at}.$$

Step 2 (integrate). Integrate from 0 to t:

$$e^{at}y(t) - e^{a\cdot 0}y(0) = \int_0^t be^{as}\,ds = \frac{b}{a}(e^{at} - 1).$$

So

$$e^{at}y(t) = y(0) + \frac{b}{a}(e^{at} - 1).$$

Step 3 (solve for $y(t)$ and apply the initial condition). With $y(0) = -1$:

$$y(t) = \left(-1 - \frac{b}{a}\right)e^{-at} + \frac{b}{a}.$$

Plugging numbers:

$$y(t) = (-7)e^{-0.5t} + \frac{3}{0.5}.$$

Quick check. As $t \to \infty$, $e^{-at} \to 0$ so $y(t) \to b/a$, the steady state of the ODE. ■

Problem 2.56 (Differential equations (second-order constant coefficients)). *Solve* $y''(t)+5y'(t)+1y(t) = 0$ *with* $y(0) = 2$ *and* $y'(0) = 0$.

Solution. This is a linear homogeneous ODE with constant coefficients. Use the characteristic equation.
Step 1 (characteristic polynomial). Assume $y(t) = e^{rt}$. Plug into the ODE:

$$r^2e^{rt} + are^{rt} + be^{rt} = 0 \quad \Rightarrow \quad r^2 + ar + b = 0.$$

Here $a = 5$ and $b = 1$, so

$$r^2 + 5r + 1 = 0.$$

Step 2 (roots). The roots are

$$r_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2}.$$

Compute the discriminant: $a^2 - 4b = 21 > 0$, so two distinct real roots:

$$r_1 \approx -0.2087, \qquad r_2 \approx -4.7913.$$

Problem 7.74 (Equity/FX vol model (Dupire local volatility: definition + pitfalls))**.** *Explain the Dupire local volatility idea: given a full surface of European call prices $C(T, K)$ (or implied vols), how is the local volatility $\sigma_{loc}(T, K)$ defined, and what are the key practical issues in using it for pricing and hedging?*

Solution. Local volatility models assume the underlying follows

$$dS_t = (r - q)S_t\, dt + \sigma_{\text{loc}}(t, S_t)S_t\, dW_t$$

under the risk-neutral measure, where the diffusion coefficient depends deterministically on time and spot level.

Definition (Dupire). If you are given a smooth arbitrage-free surface of call prices as a function of maturity and strike, $C(T, K)$, then (under standard regularity assumptions) the local variance is

$$\sigma^2_{\text{loc}}(T, K) = \frac{\partial_T C(T, K) + (r - q)K\, \partial_K C(T, K) + q\, C(T, K)}{\frac{1}{2}K^2\, \partial_{KK} C(T, K)}.$$

Intuition: $\partial_{KK} C$ encodes the risk-neutral density (via Breeden–Litzenberger), and the PDE links time evolution to diffusion strength.

Practical issues.

1) Differentiation noise. The formula requires first derivative in T and second derivative in K, which is numerically unstable unless you smooth/interpolate carefully. Garbage-in produces arbitrage and explosive local vols.

2) Static fit vs dynamic smile. Local vol is constructed to match vanilla prices at time 0 (static calibration). But it implies a specific smile dynamics (how implied vol moves as spot moves) which often does not match markets; hence hedging can be poor for exotics sensitive to smile dynamics.

3) No-arbitrage constraints. Your input surface must be arbitrage-free (monotone/convex in strike, monotone in maturity). In practice you enforce these constraints during surface construction.

4) Extrapolation. Exotics often depend on tails (very low/high strikes, long maturities) where market data is sparse. Local vol is extremely sensitive to extrapolation choices.

When to use. Local vol is popular for equity/FX exotics where vanilla calibration is required and the product is not too sensitive to volatility-of-volatility effects; for strong smile-dynamics sensitivity, stochastic volatility (e.g., Heston/SABR) or local-stochastic vol (LSV) is more appropriate.

Interview note. The interviewer usually wants: (i) definition, (ii) why it's unstable numerically, (iii) why it matches vanillas but not necessarily exotics hedges. ■

Problem 7.75 (Stochastic models (GBM solution and moments))**.** *Let S_t follow geometric Brownian motion $dS_t = \mu S_t\, dt + \sigma S_t\, dW_t$ with $S_0 = 50$. (a) Solve for S_t explicitly. (b) Compute $\mathbb{E}[S_T]$ and $\operatorname{Var}(S_T)$.*

Solution. (a) Solve the SDE by applying Itô to $\ln S_t$.

Step 1 (Itô on $\ln S_t$). For $f(s) = \ln s$, we have $f'(s) = 1/s$, $f''(s) = -1/s^2$. With $dS_t = \mu S_t\, dt + \sigma S_t\, dW_t$:

$$d(\ln S_t) = \frac{1}{S_t}\, dS_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)(\sigma^2 S_t^2)\, dt = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma\, dW_t.$$

Step 2 (integrate). Integrate from 0 to t:

$$\ln S_t - \ln S_0 = \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t.$$

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Step 2 (integrate). Integrate from 0 to t:

$$\ln S_t - \ln S_0 = \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t.$$

Exponentiate:

$$S_t = S_0 \exp\!\left(\left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W_t\right).$$

(b) Moments. Since $W_T \sim N(0, T)$, $\sigma W_T \sim N(0, \sigma^2 T)$.

Mean:

$$\mathbb{E}[S_T] = S_0 e^{\mu T}.$$

Variance:

$$\mathrm{Var}(S_T) = S_0^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right).$$

Plugging numbers ($S_0 = 50$, $\mu = 0.03$, $\sigma = 0.3$, $T = 1$):

$$\mathbb{E}[S_T] \approx 51.5227, \qquad \mathrm{Var}(S_T) \approx 249.9942.$$

Interview note. Under the risk-neutral measure for equities with dividend yield q, you replace μ by $r - q$ for pricing. ∎

Problem 7.86 (Rates/FX vol model (SABR: parameter intuition + calibration))**.** *In the SABR stochastic volatility model (common for interest-rate and FX smiles), the dynamics are*

$$dF_t = \sigma_t F_t^{\beta}\, dW_t, \qquad d\sigma_t = \nu \sigma_t\, dZ_t, \qquad d\langle W, Z\rangle_t = \rho\, dt.$$

(a) Interpret the parameters α, β, ν, ρ (at least qualitatively) and what features of the implied volatility surface they control. (b) Outline a practical calibration workflow to market implied vols for a fixed expiry.

Solution. (a) Parameter interpretation.

- $\beta \in [0, 1]$ controls how the instantaneous volatility scales with the forward level. $\beta = 1$ is lognormal (Black-style), $\beta = 0$ is normal (Bachelier-style). In rates, $\beta < 1$ is often used to accommodate low/negative rates historically (though the market has evolved).

- α is the initial volatility level (often written as σ_0). For a fixed β, changing α mostly shifts the overall level of implied volatility.

- ν is the vol-of-vol. Higher ν increases curvature (smile/smirk intensity) because the volatility itself moves more.

- ρ is the correlation between forward moves and volatility moves. It primarily controls skew: negative ρ tends to produce a left skew (implied vol higher for low strikes), positive ρ tends to produce right skew.

(b) Practical single-expiry calibration workflow.

1) Choose a convention (normal vs lognormal) and set/fit β. In many desks β is fixed by asset class policy (e.g., fixed per currency/tenor bucket) to stabilize calibration.